

Control Systems (EEE/ECE/INSTR F242)

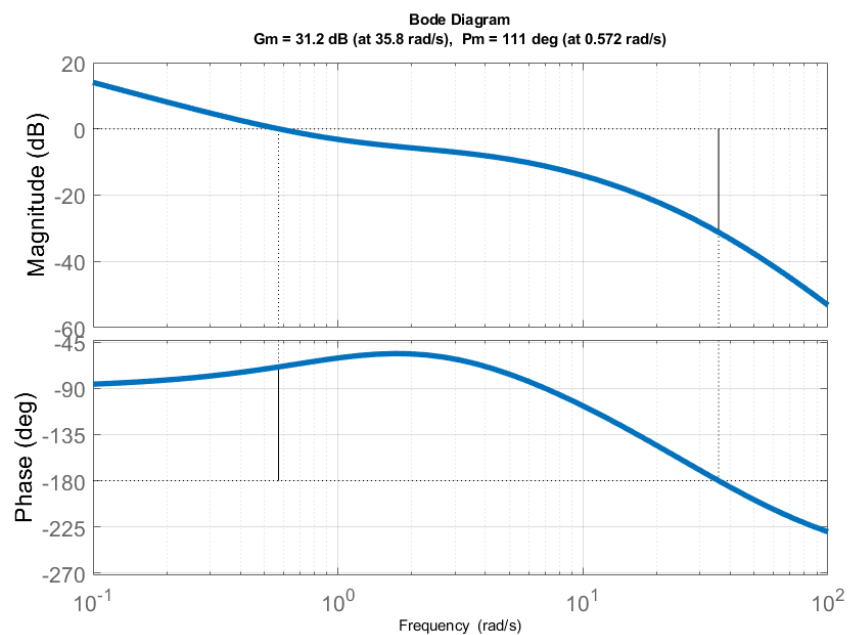
Tutorial 11 (7 April 2026)

1. For unity feedback closed loop system, the forward path transfer function $G(s)$ is given as,

$$G(s) = K \frac{(s + 1)}{s(s + 5)(s + 20)(s + 50)}$$

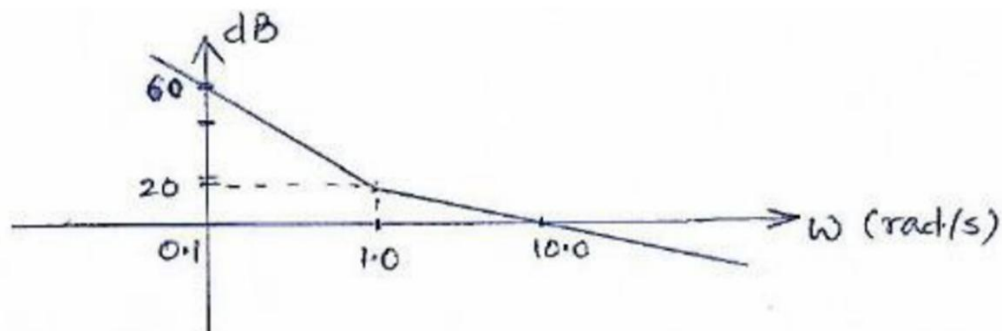
Where, $K = 2500$.

- (A) Draw the asymptotic bode plot of this system.
(B) Find the gain margin (GM) and phase margin (PM).
(C) Determine the stability of the system.
(D) What will be the effect on GM and PM, if the value of K increases 10 times.



INDICATIVE SOL.:

2. Determine the transfer function of a minimum phase system whose Bode magnitude plot is obtained experimentally (shown below).



Indicative Sol.: $G(s) = 10(s + 1)/s^2$
